

MASINASIN AGDM, AB, Canada

WMO: 713450

Lat: 49.1367N

Lon: 111.6517W

Elev: 3110

StdP: 13.12

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-16.9	-10.9	-22.9	1.7	-16.6	-16.8	2.5	-10.6	34.7	38.6	30.3	35.4	7.4	220	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	29.2	91.5	61.0	88.1	60.1	84.3	59.5	64.7	81.2	62.9	81.1	61.3	79.3	10.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	84.2	69.8	56.7	77.0	67.9	54.7	71.5	65.6	31.4	81.3	30.0	81.3	28.8	79.9	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.4	25.6	22.7	DB	-26.2	98.0	4.3	3.0	-29.2	100.2	-31.7	102.0	-34.1	103.7	-37.2	105.9	(4)
(5)			WB	-26.5	67.9	4.2	2.5	-29.5	69.7	-31.9	71.1	-34.3	72.6	-37.3	74.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	23.3	21.6	32.6	42.4	52.3	59.8	67.6	65.5	56.5	44.0	32.4	21.7	(6)	
	DBStd	19.65	15.04	15.45	13.06	9.08	7.99	6.22	5.91	6.80	8.92	9.54	13.44	14.63	(7)	
	HDD50	4166	827	794	543	254	68	4	0	1	37	227	535	876	(8)	
	HDD65	8132	1292	1214	1004	679	398	178	40	78	279	650	979	1341	(9)	
	CDD50	1772	0	0	3	24	140	298	546	480	232	43	6	0	(10)	
	CDD65	265	0	0	0	0	5	22	121	93	23	1	0	0	(11)	
	CDH74	4717	0	0	0	16	176	443	1974	1569	519	20	0	0	(12)	
	CDH80	1831	0	0	0	1	35	114	839	656	184	2	0	0	(13)	
(14) Wind	WSAvg	10.8	12.4	10.6	11.4	11.4	10.7	10.2	8.9	8.7	9.7	11.2	12.2	12.1	(14)	
(15) Precipitation	PrecAvg	14.1	0.6	0.5	0.7	1.3	1.9	3.2	1.5	1.4	1.3	0.9	0.6	0.5	(15)	
	PrecMax	22.4	1.6	1.3	1.7	2.8	3.9	6.7	5.3	4.3	3.0	3.1	1.2	1.2	(16)	
	PrecMin	8.4	0.1	0.0	0.0	0.2	0.3	0.5	0.1	0.1	0.0	0.2	0.0	0.0	(17)	
	PrecStd	3.1	0.3	0.4	0.5	0.7	0.9	1.5	1.1	1.0	0.8	0.7	0.3	0.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	56.9	65.0	76.2	84.3	89.9	96.5	94.9	90.4	77.0	65.3	53.0	(19)	
		MCWB	42.3	41.8	47.1	52.8	57.1	62.5	62.9	60.9	58.8	54.8	47.7	40.9	(20)	
	2%	DB	47.5	49.5	59.6	68.9	79.1	83.4	92.4	90.8	85.7	69.4	58.1	47.3	(21)	
		MCWB	39.0	39.2	44.5	49.5	55.6	60.4	62.2	59.9	57.2	50.9	45.4	39.1	(22)	
	5%	DB	44.1	44.8	55.1	63.8	74.2	79.2	88.6	87.5	80.5	63.8	53.4	43.5	(23)	
		MCWB	37.0	36.3	42.3	46.5	53.4	59.0	61.6	58.9	55.8	48.8	42.7	36.5	(24)	
	10%	DB	41.1	41.1	50.5	59.0	69.6	75.4	84.7	83.2	74.6	59.2	49.0	40.2	(25)	
		MCWB	35.2	34.7	40.0	44.2	51.7	57.0	60.9	58.5	54.0	46.3	40.0	34.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	43.4	48.4	53.1	58.3	65.9	68.2	64.6	61.6	56.3	50.1	43.1	(27)	
		MCDB	50.5	54.1	63.1	72.9	79.7	80.5	86.9	80.3	80.1	73.0	63.0	51.0	(28)	
	2%	WB	39.7	39.6	45.5	50.1	56.2	62.9	65.8	62.5	58.9	52.4	45.9	39.6	(29)	
		MCDB	46.8	48.7	57.5	67.6	75.4	79.8	81.5	83.3	79.6	66.7	57.1	46.6	(30)	
	5%	WB	37.4	37.0	43.0	47.3	54.5	60.4	63.9	61.0	56.8	49.4	43.1	36.8	(31)	
		MCDB	43.2	44.2	54.2	62.0	71.3	75.5	81.3	81.2	77.3	61.6	52.4	42.8	(32)	
	10%	WB	35.5	34.8	40.3	44.7	52.7	58.4	62.4	59.6	54.8	47.1	40.5	34.6	(33)	
		MCDB	40.9	40.7	49.8	57.3	67.0	72.5	80.5	78.8	73.9	57.9	48.5	40.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	20.0	21.2	23.1	23.8	23.8	29.2	28.5	25.8	22.6	19.7	17.8	(35)	
		MCDBR	18.5	23.7	27.7	32.2	32.3	31.2	35.9	36.5	36.2	29.4	23.8	18.9	(36)	
	5% WB	MCWBR	12.3	15.1	15.5	15.8	13.6	12.9	12.7	12.9	14.6	14.7	13.6	13.0	(37)	
		MCWBR	17.9	22.3	26.3	30.6	28.7	28.2	29.7	30.9	32.8	27.6	23.3	18.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.255	0.263	0.282	0.327	0.343	0.340	0.342	0.362	0.322	0.285	0.258	0.247	(40)	
		taud	2.463	2.494	2.499	2.371	2.370	2.431	2.436	2.351	2.435	2.527	2.510	2.446	(41)	
	Ebn at Noon		264	289	299	292	291	292	289	277	279	275	261	251	(42)	
		Edh at Noon	19	23	28	35	37	35	34	36	30	23	18	17	(43)	
(44) All-Sky Solar Radiation	RadAvg		406	715	1121	1554	1849	2067	2256	1816	1294	809	459	319	(44)	
	RadStd		34	59	89	99	144	127	118	113	102	77	32	29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page